How N-HiTS makes a forecast

Challu et al. (2023) · supervised baseline trained on each dataset · Darts NHiTSModel

1. A window of inputs

Each forecast reads a window of the last 400 days (36 months). A ? is a value not known yet.

past *future*

Target temperature

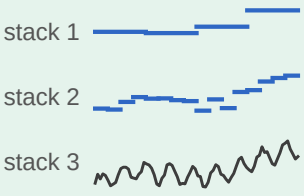
Past only NAO index

Prices go in as log differences, oil and yields as differences; all inputs are min-max scaled on the training rows.

N-HiTS takes past-only covariates; day length is not used.

2. Three stacks, three time scales

Each stack first max-pools the window: the first sees only the broad shape, the last every step.



Daily pooling sizes 200, 14 and 1 steps (monthly 18, 4, 1).

3. A few points, then interpolate

In each stack an MLP predicts only a few forecast points, and straight lines join them up to every step.

few points → interpolated forecast

MLP: 2 layers, width 512 (tuned 128 to 512), ReLU

Points per stack: 2, 15 and 90 for 90 days; each stack also returns the part of the window it explained, which is removed before the next

4. Add the stacks, train once

Trained once per dataset on the rows before the first test window (daily: the last 4,096), then every window is forecast from its own origin with no retraining.

point forecast

The forecast is the sum of the three stacks. Optuna tries a few settings on validation windows inside the training rows; the first is the default. 10 epochs, MSE loss.

Coarse stacks capture the slow movements cheaply; the last stack fills in the detail.

3 stacks of 1 block · 2 layers each · window 400 days or 36 months · direct forecast of 90 days or 24 months · tuned: width, learning rate